

Spatial-Temporal Mixture-of-Graph-Experts for Multi-Type Crime Prediction

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Abstract As various types of crimes continue to threaten public safety and economic development, predicting the occurrence of multiple types of crimes becomes increasingly vital for effective prevention measures. Although extensive efforts have been made, most of them overlook the heterogeneity of different crime categories and fail to address the issue of imbalanced spatial distribution. In this work, we propose Spatial-Temporal Mixture-of-Graph-Experts (ST-MoGE), a framework for collective multiple-type crime prediction. To enhance the model's ability to identify diverse spatial-temporal dependencies and mitigate potential conflicts caused by spatial-temporal heterogeneity of different crime categories, we introduce a module, Mixture-of-Graph-Experts (MGE), to capture the distinctive and shared crime patterns of each crime category. Then, we propose Cross-Expert Contrastive Learning (CECL) to refine MGE and force each expert to specialize in modeling specific patterns, thereby reducing blending and redundancy. Furthermore, to address the issue of imbalanced spatial distribution, we propose a module, Hierarchical Adaptive Loss Re-Weighting (HALR), to eliminate biases and underfitting in data-scarce regions. To evaluate the effectiveness of our methods, we conduct comprehensive experiments on two real-world crime datasets and compare our results with 12 advanced baselines. The experimental results demonstrate the superiority of our methods.

Keywords multi-type crime prediction, spatio-temporal prediction, mixture-of-experts

1 Introduction

As various crimes (e.g., theft, assault, robbery) continue to rise, they have become a significant threat to public safety and economic development^[1]. Contrary to the intuitive belief that criminal acts are ran-

dom and unpredictable, crime pattern theory suggests that crimes are either planned or opportunistic, following implicit patterns^[2]. Accurate forecasting of crime occurrences thus becomes a cornerstone for effective crime prevention. By enabling proactive strategies, crime prediction plays a pivotal role in ur-

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ban governance and public safety.

Due to its importance, numerous machine learning based crime prediction methods have recently been developed, generally categorized into deep attentive based and graph-based approaches. Deep attentive based approaches^[3, 4] aim to model crime dynamics using attention mechanisms. On the other hand, graph-based approaches^[5–8] aim to capture spatial-temporal relationships among different crime patterns. However, these methods still struggle to effectively model spatial-temporal dependencies among different crime categories due to their inherent spatial-temporal heterogeneity.

Spatial-temporal heterogeneity arises from distinct patterns unique to each crime category. For instance, burglaries are intuitively more likely to occur in residential zones, whereas larcenies prevail in commercial areas. Fig.1 illustrates the distinct spatial distributions of different crime categories, highlighting their pattern heterogeneity. Traditional spatial-temporal models for crime prediction often capture only shared patterns across categories, overlooking category-specific characteristics. This limitation gives rise to two key issues. 1) The unique crime patterns, especially those that are distinctive, cannot be fully captured, resulting in incomplete pattern modeling. 2) Crime patterns across categories may conflict in certain regions, making accurate modeling difficult^[9]. Although predicting each category separately may avoid such conflicts, we argue that a unified multi-type framework brings distinct advantages. First, joint modeling better captures shared spatial-temporal patterns across crime categories. Second, a unified framework is more scalable and computationally efficient than managing separate models. Finally, multi-type learning enables knowledge transfer from frequent to rare crimes, mitigating data sparsity and improving predictions for low-frequency types. Nonetheless, capturing heterogeneous patterns across crime categories

remains a major challenge, as increasing model size or tailoring architectures alone is insufficient. Addressing this requires a model that both accommodates pattern heterogeneity and mitigates inter-category conflicts.

Recently, the Mixture-of-Experts (MoE) architecture has demonstrated its superiority in handling heterogeneous data due to its remarkable adaptability and generalization capabilities^[10]. However, directly applying MoE to multi-type crime prediction is non-trivial. First, there is the challenge of dynamically routing experts to integrate spatial-temporal knowledge. Most existing MoE models target static tasks with relatively stable patterns, whereas crime prediction involves highly non-stationary and volatile spatial-temporal dynamics. This variability requires flexible expert assignment that adapts to changing conditions. For instance, property crimes may spike in residential areas at night, while violent crimes increase in commercial zones on weekends. Therefore, MoE for crime prediction must route appropriate experts for different crime categories, adapting to changes in spatial and temporal crime distributions, which is challenging. Second, the spatial imbalance of crime data increases the difficulty. As shown in Fig.2, crime occurrences follow a long-tailed distribution, concentrated in a few regions. Traditional MoE frameworks can be easily influenced by these regions, harming performance in other regions. Therefore, avoiding such bias presents another significant challenge. These challenges necessitate a sophisticated mechanism to ensure that the MoE framework can adapt to the dynamic and heterogeneous nature of crime data.

To address this, we propose a new spatio-temporal crime prediction framework, named Spatial-Temporal Mixture-of-Graph-Experts (ST-MoGE). Specifically, we introduce the Mixture-of-Graph-Experts (MGE) module, which comprises multiple spatial-temporal graph learning experts—both crime-specific and universal experts—to fully capture heterogeneous and universal crime patterns. An attentive gating mechanism dynamically routes specific experts for the selective integration of extracted spatial-temporal knowledge. Additionally, to ensure that optimal crime-dependent experts are routed and to decompose the learning of unique crime patterns in parallel, we propose Cross-Expert Contrastive Learning (CECL) to ensure that each crime-dependent expert focuses on learning its specific crime pattern. Furthermore, to address the challenge of imbalanced spatial distribution, we propose regional-aware predictors. These pre-

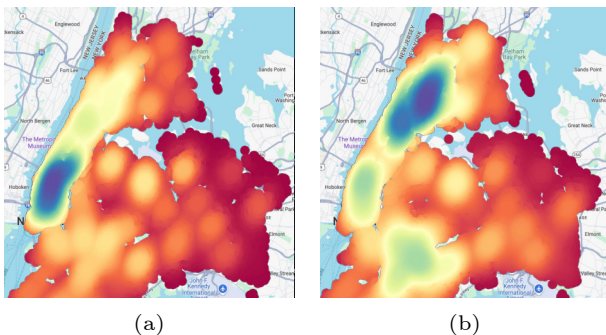


Fig.1. Spatial distribution of crimes in New York City. (a) Larceny. (b) Harassment.

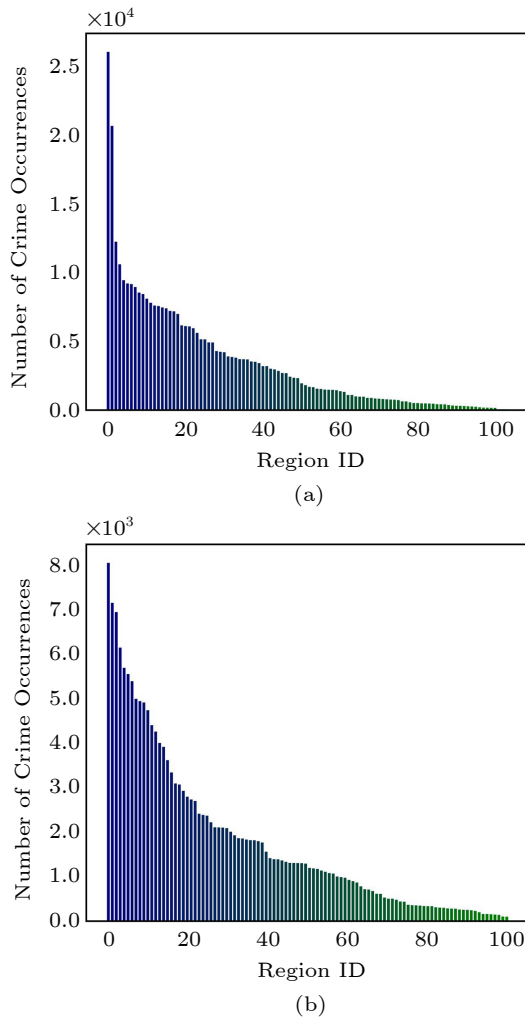


Fig.2. Crime occurrence distribution across regions in New York City. (a) Larceny. (b) Harassment.

dictors dynamically adjust weights across different regions to mitigate disproportionate influences during model training.

The key contributions can be summarized as follows.

- We present ST-MoGE, a tailored framework for collective multi-type crime prediction. To the best of our knowledge, this is the first attempt to address the spatial-temporal heterogeneity of different crime categories using an MoE architecture.
- We propose an MoE-architected spatial-temporal network that comprehensively captures heterogeneous spatial-temporal characteristics. We further devise a contrastive learning approach, relieving the redundancy and enhancing the concentration of each expert. Moreover, we devise a module, Hierarchical Adaptive Loss Re-Weighting (HALR), to relieve the training biases caused by the imbalanced spatial dis-

tribution of crimes.

- We conduct comprehensive experiments on two real-world crime datasets from New York City (NYC) and Chicago (CHI), respectively, thoroughly evaluating 12 spatial-temporal prediction methods from various angles. The experimental results demonstrate ST-MoGE's superior performance.

The remainder of this paper is organized as follows. [Section 2](#) reviews relevant literature. [Section 3](#) introduces preliminary concepts and definitions. In [Section 4](#), we describe the design and components of the proposed ST-MoGE. [Section 5](#) presents experiments and results. Finally, [Section 6](#) concludes the paper.

2 Related Work

2.1 Spatial-Temporal Prediction

Spatial-temporal prediction involves analyzing and predicting the evolution of state or value across both spatial and temporal dimensions. It has been widely used for tasks of urban computing, such as traffic flow prediction^[11] and public transportation optimization^[12]. Extensive methodologies are proposed for spatial-temporal prediction. Early methodologies employ statistical and conventional machine learning models, such as AutoRegressive Integrated Moving Average (ARIMA), Support Vector Regression (SVR), and Random Forest Regression^[13]. In recent years, deep learning methods have shown remarkable performance in various urban computing tasks involving spatial-temporal prediction^[14–17]. Time-series prediction methods such as recurrent neural networks (RNNs) and temporal convolutional networks (TCNs) are widely applied for modeling temporal dependencies. RNNs are applied in models such as Spatial Temporal Recurrent Neural Networks (ST-RNN)^[18]. On the other hand, TCNs are leveraged in Graph WaveNet (GWN)^[19] and Multivariate Time-series Graph Neural Network (MTGNN)^[20], demonstrating their effectiveness in handling temporal dynamics. To capture spatial dependencies, convolutional neural networks (CNNs) are often applied to grid-structured data^[8]. More recently, graph neural networks (GNNs) have been adopted to model spatial dependencies in non-Euclidean structured spatial-temporal data. For instance, several methods leverage spectral graphs, such as Spatio-Temporal Graph Convolutional Network (STGCN)^[5] and Diffusion Convolutional Recurrent Neural Network (DCRNN)^[21]. To account for dy-

dynamic spatial dependencies, graph attention neural networks have been introduced to dynamically calculate region-specific importance, as seen in Graph Multi-Attention Network (GMAN)^[20]. These models do not require predefined spatial information, allowing them to discover spatial correlations in a data-driven manner. However, the aforementioned methods could be limited when applied to crime prediction due to several imbalanced data distributions and issues with multi-objective prediction.

2.2 Crime Prediction

Crime prediction has been widely researched over the years. Early approaches primarily rely on statistical and conventional machine learning models^[22]. In recent years, deep learning methods have gained significant attention in crime prediction.

Some researchers use external information to enhance prediction performance. For example, Huang *et al.*^[4] developed DeepCrime, which builds static region embeddings with Point of Interest (POI) data to capture regional traits and uses hierarchical Gated Recurrent Units (GRUs) for prediction. AttenCrime^[23] incorporates external spatial-temporal data such as anomaly data and taxi data, employing attention mechanisms to model these cross-domain relationships.

Data sparsity is a key challenge in crime prediction, limiting the model's ability to detect patterns in areas with few crime occurrences. In ^[24], an unsupervised domain adaptation method is introduced for cross-city crime risk prediction. Li *et al.*^[8] used self-supervised learning for data augmentation in Spatial-Temporal Hypergraph Self-Supervised Learning (ST-HSL), incorporating a hypergraph convolution network for crime pattern modeling. Fine-grained crime prediction worsens data sparsity due to the increased detail. In STtrans^[25], a shared embedding space for pattern extraction and adversarial training help address this issue. Zhao *et al.*^[23] proposed a classification-labeled continuousization strategy to convert sparse crime data into continuous signals, easing the sparsity problem.

Unlike deep learning methods, some studies adopt mathematical models for crime pattern modeling, offering faster inference and better interpretability. For example, Zhao *et al.*^[26] proposed a tensor factorization method for single-category prediction, later extending it to multi-category settings by modeling spa-

tial, temporal, and categorical dependencies^[27].

2.3 Mixture-of-Experts

MOE is a machine learning architecture that combines specialized sub-models, each targeting specific data subsets or tasks^[10]. It has shown strong performance in handling heterogeneous data and complex problems across various domains, including spatial-temporal prediction. For example, GESME-Net^[28] employs a spatial-temporal MoE architecture with CNN and RNN experts to address source heterogeneity in multi-city ride-hailing demand prediction. Similarly, Liu *et al.*^[29] proposed ST-MoE, which encodes spatial-temporal knowledge into embeddings and uses MLP (multi-layer perceptron)-based experts for prediction. CP-MoE^[30] employs adaptive graph learners and specialized experts to model diverse spatial-temporal traffic patterns. An ordinal regression strategy guides expert collaboration. More recently, Zhang *et al.*^[31] proposed a graph-based MoE framework for urban congestion, with experts modeling intra- and inter-mobility patterns. Distinct from existing approaches, our model employs cross-expert contrastive learning to enforce role specialization of experts and introduces an attentive spatial gate for fine-grained expert routing.

3 Preliminaries and Problem Definition

In this section, we introduce the definitions in this paper and formally formulate the research problem.

Definition 1 (Region Graph $\mathcal{G}$). *In this study, we represent the structure of a city as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$. The city is divided into N regions, with each region labeled as $R = \{r_i | r_i \in r_1, \dots, r_n, \dots, r_N\}$. Each region serves as a node in $\mathcal{V}$ and the set $\mathcal{E}$ represents a collection of edges. If two regions are adjacent geographically, their corresponding nodes in the graph are considered connected. $\mathcal{A}$ represents the adjacency matrix, defined as $\mathcal{A} \in \mathbb{R}^{N \times N}$, where*

$$\mathcal{A}_{i,j} = \begin{cases} 1, & \text{if } r_i \text{ and } r_j \text{ are connected,} \\ 0, & \text{otherwise.} \end{cases}$$

Definition 2 (Crime Tensor). *The raw crime data comprises reports of crime occurrences. Each report includes details such as the category of crime, precise timestamp, and geographical coordinates. By associating each crime occurrence with its respective time slot and region, we construct the crime tensor $\mathbf{X} \in \mathbb{R}^{N \times T \times C}$,*

where N , T , and C represent the number of regions, time slots, and crime categories, respectively. Within the crime tensor, each entry $X_{n, t, c}$ denotes the occurrence amount of a crime in category c within region r during time slot t .

Problem Formalization. Given the historical crime tensor $\mathbf{X} \in \mathbb{R}^{N \times T \times C}$ from previous T time slots and the region graph $\mathcal{G}$, the crime prediction problem aims to learn a mapping function $\mathcal{F}(\cdot)$ to predict the crime occurrence of each category in each region during the next time slot $T + 1$,

$$\mathcal{F} : (\mathbf{X}_T, \mathcal{G}) \mapsto \hat{\mathbf{X}}_{T+1} \in \mathbb{R}^{N \times C},$$

where $\hat{\mathbf{X}}_{T+1}$ represents the estimated crime tensor in time slot $T + 1$, considered as the prediction results.

4 Methods

Fig.3 presents an overview of the ST-MoGE framework, illustrating the architecture of MGE. It contains multiple spatial-temporal graph learning experts (ST-experts), including category-specific ST-experts and universal ST-experts, to capture the shared and category-specific crime patterns, respectively. Spatial attentive gates are introduced to selectively route the shared spatial-temporal knowledge of each region for different crime categories. The regional-aware predictor is further applied, in which the regions are clustered based on their spatial-temporal characteristics, and tailored predictors are deployed to

obtain the prediction results.

To enhance the prediction capacity of MGE, two training strategies are applied. 1) *Cross-Expert Contrastive Learning (CECL)*. It employs an auxiliary expert shared parameter with the universal expert, generating corrupted representations for contrasting with other experts' representations, forcing each expert to focus on its target crime pattern. 2) *Hierarchical Adaptive Loss Re-Weighting (HALR)*. It is introduced to dynamically adjust the loss weight of each predictor during the training phase, alleviating issues related to insufficient training in specific regions.

4.1 Mixture-of-Graph-Experts

Due to heterogeneous patterns across crime categories, modeling only shared spatial-temporal dependencies is insufficient. To address this issue, we propose the MGE module, which includes a universal expert for shared pattern capturing and multiple category-specific experts for distinct pattern extraction. Spatial attentive gates are applied to selectively route regions to appropriate experts.

4.1.1 Spatial-Temporal Graph Learning Expert

The spatial-temporal graph learning expert serves as a key component in MGE, aiming at modeling both spatial and temporal dependencies simultaneously. The category-specific and universal ST-experts

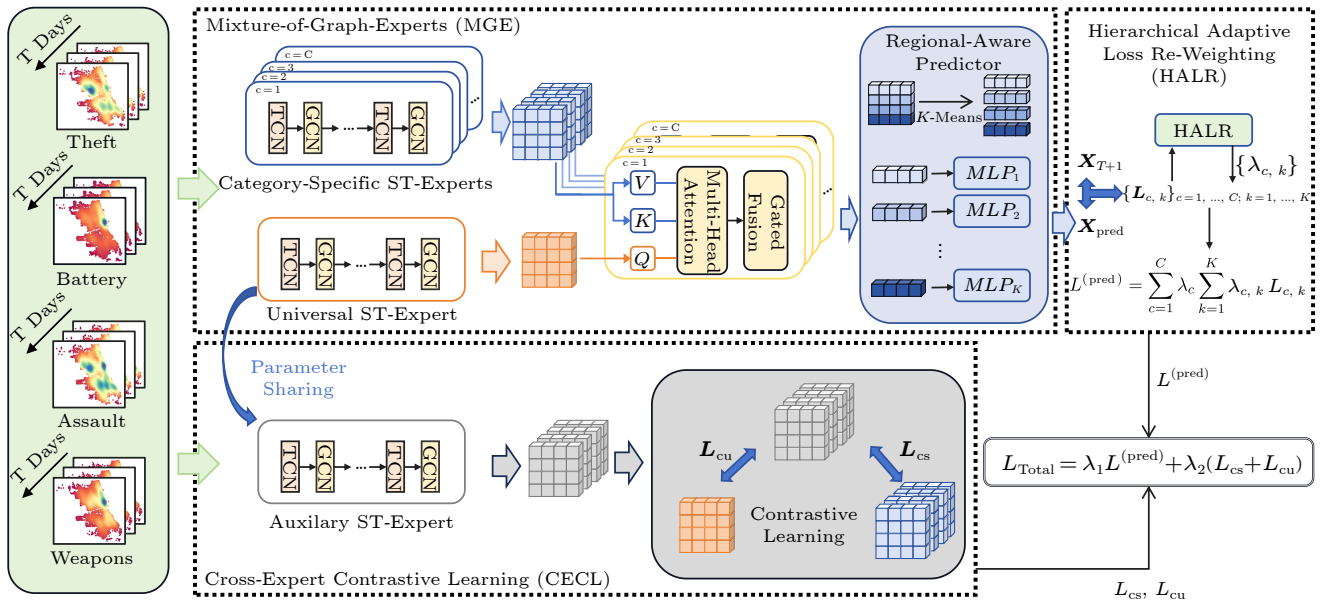


Fig.3. Architecture of the proposed ST-MoGE framework. V: values. K: keys. Q: queries.

share the same architecture but differ in their data inputs. The category-specific ST-experts receive crime tensors corresponding to their specific categories, allowing their corresponding specific categories to model distinctive crime patterns unique to each category. In contrast, the universal ST-expert processes the combined crime tensors from all categories, aiming to capture the shared crime patterns that are common across different types of crimes. Each expert consists of several stacked spatial-temporal blocks (ST-blocks), which employ graph convolutional networks (GCN) with self-adaptive adjacency matrices to autonomously capture geographical interrelations and utilize dilated temporal convolutional networks (TCN) to capture temporal patterns. The implementation details of the expert modules are provided in Appendix A1^①. After passing through the ST-experts, we obtain spatio-temporal representations that encode either category-specific or shared crime patterns. Specifically, the outputs of the category-specific and universal ST-experts are denoted as $\mathbf{H}^{(S)}$ and $\mathbf{H}^{(U)}$, respectively.

4.1.2 Attentive Spatial Gates

For different crime categories, there exists a substantial discrepancy in spatial-temporal patterns across regions. This discrepancy indicates that the need for shared knowledge from the universal expert can vary significantly by region for different crime categories. Simply concatenating the knowledge from category-specific experts and the universal expert may introduce noise or even mutually contradictory knowledge, complicating the accurate modeling of spatial-temporal patterns. To address this issue, we introduce attentive spatial gates for selective knowledge integration from diverse experts. The architecture is shown in Fig.4.

In particular, we employ a multi-head cross-attention mechanism, which identifies the relative significance of each region's representations from the universal expert. Specifically, for category c of crime, given the representation $\mathbf{H}_c^{(S)} \in \mathbb{R}^{N \times d}$ from its category-specific expert, alongside the representation $\mathbf{H}^{(U)} \in \mathbb{R}^{N \times d}$ from the universal expert, the multi-head attention mechanism conducts selectively aggregation with attention scores derived from them. The computation for the m -th attention head is articulated as

$$\hat{\mathbf{H}}_c^{(U), m} = \text{SoftMax} \left(\frac{\mathbf{W}_Q \cdot \mathbf{H}_c^{(S)} (\mathbf{W}_K \cdot \mathbf{H}^{(U)})^T}{\sqrt{d_Q}} \right) \mathbf{W}_V \cdot \mathbf{H}^{(U)},$$

where the matrices $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$ serve as trainable parameters for queries, keys, and values in the attention model, respectively. The term d_Q denotes the embedding dimension of $\mathbf{W}_Q$, and the normalization term $\sqrt{d_Q}$ is applied to mitigate overly large inner product values. The representations from each attention head are then merged, described as

$$\hat{\mathbf{H}}_c^{(U)} = \mathbf{W}^* \cdot \text{concat}(\hat{\mathbf{H}}_c^{(U), 1}, \dots, \hat{\mathbf{H}}_c^{(U), m}, \dots, \hat{\mathbf{H}}_c^{(U), M}),$$

where $\mathbf{W}^*$ is a trainable parameter that combines the outputs of multiple attention heads into a single representation. A gated fusion operation is further applied to adaptively integrate the recalibrated universal representation $\hat{\mathbf{H}}_c^{(U)}$ with the category-specific representation $\mathbf{H}_c^{(S)}$ in the following manner:

$$\mathbf{H}_c^{(F)} = \mathbf{z} \odot \mathbf{H}_c^{(S)} + (1 - \mathbf{z}) \odot \hat{\mathbf{H}}_c^{(U)},$$

$$\mathbf{z} = \sigma(\mathbf{H}_c^{(S)} \mathbf{W}_1 + \hat{\mathbf{H}}_c^{(U)} \mathbf{W}_2 + \mathbf{b}_z),$$

where $\mathbf{W}_1$, $\mathbf{W}_2$, and $\mathbf{b}_z$ denote the trainable parameters of the model. The operation $\odot$ indicates element-wise multiplication, with $\sigma(\cdot)$ being the sigmoid function, acting as a gating mechanism to modulate the

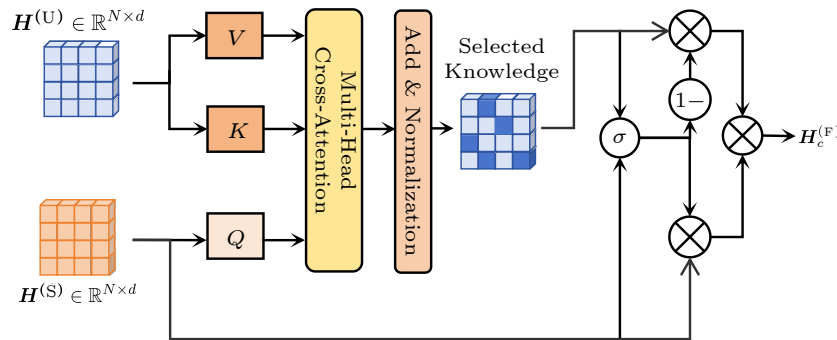


Fig.4. Structure of the attentive spatial gate. V: values. K: keys. Q: queries.

^①<https://jcst.ict.ac.cn/en/supplement/0ec730a7-5099-41ec-9951-9b2225a50bcc>, Mar. 2026.

integration from each expert.

4.1.3 Regional-Aware Predictor

Existing crime prediction approaches usually employ a predictor with shared parameters to predict crimes across all regions. However, the skewed crime distribution can introduce substantial bias if a single shared predictor is utilized. Therefore, it is essential to learn a predictor with a separate set of parameters for regions and crime categories with different spatial-temporal characteristics.

To achieve this, we leverage a clustering-based mechanism to encourage regions with similar spatial-temporal characteristics to form clusters. The node embedding E learned in each category-specific ST-expert preserves the unique parameters of each region and reflects their spatial-temporal characteristics. For each crime category, we cluster the regions into K clusters with the K -means algorithm^[32]. Consequently, the corresponding representations of each cluster are deployed with an MLP as the tailored predictor to obtain the prediction results of the involved regions. Formally, the process can be represented as follows:

$$\hat{\mathbf{X}}_{c, k} = MLP_k^c(\mathbf{H}_{c, k}^{(F)}),$$

where $\mathbf{H}_{c, k}^{(F)}$ represents the selected representation of region cluster k , crime category c , and $\hat{\mathbf{X}}_{c, k}$ denotes the prediction result for this cluster. The integrated prediction result is achieved by aggregating the results from each cluster.

4.2 Training Strategies

Although the ST-MoGE network is adept at capturing spatial-temporal dependencies inherent in criminal incidents, it is negatively influenced by challenges such as cross-expert information blending and the imbalanced distribution of crimes. In order to address these challenges and enhance the model's capacity, we propose the implementation of the following training strategies: CECL and HALR.

4.2.1 Cross-Expert Contrastive Learning

Though in the MGE module, multiple experts are employed to capture different crime patterns, without any constraints on the differentiation of target patterns, there may exist blendings across experts in the implementation. Such blending could result in non-eligible redundancy or even contradictory knowl-

edge. To mitigate this issue, we introduce cross-expert contrastive learning, which decomposes the extracted distinctive and mutual patterns in a self-supervised method.

The key insight of CECL is to force the representations encoded with different types of crime patterns to be far away from each other in the latent space. However, since the input for different experts is inherently different, directly leveraging the original representations to construct negative pairs yields minimal benefit. Therefore, an auxiliary ST-expert is deployed for corrupted representation generation. This expert shares parameters with the universal expert. The corrupted representation of the c -th crime category, denoted as $\tilde{\mathbf{H}}_c$, can be generated by inputting crime data of the corresponding category. Our objective is to maximize the distinction between corrupted representations and ordinary representations. Hence, the negative pairs can be constructed as a corrupted representation with representations from the corresponding category-specific expert or the universal expert. Moreover, inspired by SimCSE^[33], the representation for positive pairs construction, denoted as $\tilde{\mathbf{H}}'_c$, is generated in a similar approach, with small disturbance by dropout operations. The InfoNCE^[34] loss function is adopted, aiming to maximize the lower bound on mutual information for positive pairs while minimizing it for negative pairs. The contrastive loss L_{cs} for category-specific experts and L_{cu} for the universal expert can be expressed as

$$L_{cs} = -\frac{1}{C} \sum_{c=1}^C \log \frac{\exp(\text{sim}(\tilde{\mathbf{H}}'_c, \tilde{\mathbf{H}}_c)/\tau)}{\sum_{c=1}^C \exp(\text{sim}(\mathbf{H}_c^{(S)}, \tilde{\mathbf{H}}_c)/\tau)}, \quad (1)$$

$$L_{cu} = -\frac{1}{C} \sum_{c=1}^C \log \frac{\exp(\text{sim}(\tilde{\mathbf{H}}'_c, \tilde{\mathbf{H}}_c)/\tau)}{\sum_{c=1}^C \exp(\text{sim}(\mathbf{H}^{(U)}, \tilde{\mathbf{H}}_c)/\tau)}. \quad (2)$$

Here, τ represents the temperature parameter, and $\text{sim}(\cdot)$ denotes the cosine similarity function. By applying CECL, the extracted knowledge of different experts can be efficiently separated, consequently decomposing the distinctive and shared crime patterns and relieving the blending among experts.

4.2.2 Hierarchical Adaptive Loss Re-Weighting

Real-world crime activities normally follow a long-tailed spatial distribution. In the training stage, the model is easily insufficiently trained on regions with

relatively low crime frequency due to the skewed crime distribution. A common approach to mitigating this issue is static loss re-weighting based on crime frequency. However, such methods often lead to instability in extremely sparse areas. To address this, we propose the HALR module that dynamically adjusts the training focus based on real-time learning progress. Unlike static frequency-based weighting, our method monitors the rate of loss reduction and increases the attention to region clusters and categories where learning is lagging. The weighting coefficient for category c , $\lambda_c(t)$, is defined as

$$\lambda_c(t) = \frac{C \exp(\mathbf{w}_c(t-1)/T)}{\sum_{c=1}^C \exp(\mathbf{w}_c(t-1)/T)},$$

with

$$\mathbf{w}_c(t-1) = \frac{L_c(t-1)}{L_c(t-2)}, \quad (3)$$

where $\mathbf{w}_c(\cdot)$ measures the rate of loss reduction, t indicates the iteration step, and T denotes a tuning parameter that adjusts the smoothness of weight distribution. A larger T value can promote a more equitable weight distribution. The term L_c represents the error in predictions for category c .

For clusters belonging to the same crime category, we calculate the coefficient employing a similar method by replacing L_c in (3) with $L_{c,k}$, which indicates the prediction error for cluster k within category c . Consequently, the re-weighted loss is depicted as

$$L^{(\text{pred})} = \sum_{c=1}^C \lambda_c(t) \sum_{k=1}^K \lambda_{c,k}(t) \|\mathbf{X}_{c,k}^{T+1} - \hat{\mathbf{X}}_{c,k}(t)\|_2^2, \quad (4)$$

where $\mathbf{X}_{c,k}^{T+1}$ and $\hat{\mathbf{X}}_{c,k}$ denote the ground truth and prediction results for cluster k and category c , respectively.

4.3 Joint Optimization

The composite loss function integrates three key elements. The first element is the re-weighted prediction loss $L^{(\text{pred})}$, calculated in (4). The second element contains the contrastive losses L_{cs} and L_{cu} , obtained in (1) and (2), respectively.

Overall, the ST-MoGE model is trained by jointly optimizing the following objective function:

$$L = \lambda_1 L^{(\text{pred})} + \lambda_2 (L_{\text{cs}} + L_{\text{cu}}).$$

In this formulation, λ_1 and λ_2 act as the regulatory coefficients that optimize the balance of losses, collectively summing to 1.

A detailed breakdown of the computational complexity of each module is provided in Appendix A2^②.

5 Experiments

To validate the effectiveness and generalizability of ST-MoGE, we perform extensive evaluations focusing on overall performance, module contributions, robustness under different hyperparameter settings, and performance across spatial and categorical crime sparsity.

5.1 Experimental Setup

5.1.1 Datasets

We evaluate our proposed ST-MoGE framework in two real-world crime datasets, the NYC dataset and the CHI dataset, collected from New York City (NYC) and Chicago (CHI), respectively. NYC and CHI are divided into 225 and 140 disjoint regions, respectively. The adjacency matrix of these regions is constructed based on the geographical proximity relationship between regions. The time slot is set to be 1 day. T is set to 7, which means that data from 7 days before the target day is used as the input to the model. The training, validation, and test datasets are generated by partitioning both datasets with a ratio of 8:1:1.

5.1.2 Evaluation Metrics

To assess the performance of crime prediction, we employ the mean absolute error (MAE) and mean absolute percentage error (MAPE) as the evaluation metrics. MAE measures the average magnitude of errors between actual and predicted values. MAPE calculates the average percentage difference between actual and predicted values, providing a relative measure of accuracy that is easy to interpret.

We adopt these two metrics based on the nature of our task, which focuses on estimating the magnitude of crime counts rather than merely detecting

^②<https://jcst.ict.ac.cn/en/supplement/0ec730a7-5099-41ec-9951-9b2225a50bcc>, Mar. 2026.

their presence or absence. MAE offers a robust and interpretable measure of absolute error, whereas MAPE, being scale-independent, facilitates comparison across regions with different crime frequencies. This is particularly important in real-world scenarios such as patrol resource allocation and hotspot prioritization, where understanding the expected crime volume (e.g., 2 vs 5 incidents) is more actionable than simply predicting whether a crime will occur or not.

These metrics are defined as follows:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|,$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|,$$

where y_i is the actual value, $\hat{y}_i$ represents the predicted value, and n is the total number of samples.

5.1.3 Baselines

We compare the ST-MoGE framework against 12 baseline methods for crime prediction, which are grouped into three categories: 1) GNN-based prediction methods: STGCN, GWN, GMAN, MTGNN, AGCRN (Adaptive Graph Convolutional Recurrent Network), MAGNN (Multi-Scale Adaptive Graph Neural Network), and FCSTGCN (Fully Connected

Spatio-Temporal Graph Convolutional Network); 2) crime prediction methods: DeepCrime, ST-HSL, and STSHN (Spatial-Temporal Sequential Hypergraph Network); and 3) mixture-of-experts methods: STGCN-MoE and ST-MoE.

Detailed configurations, including dataset statistics, hyperparameter settings, implementation details, and baseline descriptions, are provided in Appendix A3^③.

5.2 Performance Comparison

Tables 1 and 2 report the performance comparison results between our ST-MoGE framework and various baseline models for crime prediction. In all the results tables, the best results are highlighted in bold, and MAE and MAPE are negatively oriented metrics, where smaller values correspond to higher predictive accuracy. We summarize our findings as follows.

- ST-MoGE outperforms all the compared baseline models on all the crime categories within both crime datasets. We attribute such improvements to the following reasons. 1) By leveraging the MoE architecture network, ST-MoGE could comprehensively preserve heterogeneous crime patterns of different categories. 2) Benefiting from contrastive learning, ST-MoGE is able to decompose different types of crime patterns, reducing the modeling pattern information

Table 1. Overall Performance (MAE and MAPE) of Crime Prediction on the NYC Dataset

Model	Larceny		Harassment		Assault		Mischief		Indecency		Robbery		Overall	
	MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓
STGCN ^[5]	1.6206	0.6030	0.9722	0.5649	0.9631	0.5495	0.6654	0.5739	0.3428	0.7353	0.3597	0.7216	0.8206	0.6247
GWN ^[5]	1.4901	0.5627	0.9618	0.5851	0.9490	0.5618	0.6541	0.5814	0.3414	0.7697	0.3594	0.7349	0.7926	0.6326
GMAN ^[7]	1.5748	0.5284	0.9577	0.5779	0.9404	0.5828	0.6463	0.5815	0.3390	0.7193	0.3523	0.6734	0.8018	0.6106
MTGNN ^[20]	1.5807	0.5254	0.9758	0.5900	0.9541	0.5763	0.6570	0.5966	0.3350	0.7562	0.3620	0.7277	0.8108	0.6287
AGCRN ^[6]	1.5844	0.5328	0.9544	0.5592	0.9409	0.5548	0.6314	0.5662	0.3282	0.7117	0.3593	0.7126	0.7998	0.6062
MAGNN ^[35]	1.5189	0.5366	0.9921	0.5699	0.9700	0.5563	0.6608	0.5935	0.3553	0.7039	0.3823	0.6594	0.8132	0.6033
FCSTGNN ^[36]	1.5679	0.5500	0.9689	0.5989	0.9570	0.6027	0.6514	0.6367	0.3387	0.7525	0.3591	0.7510	0.8072	0.6486
DeepCrime ^[4]	1.6917	0.5273	0.9511	0.5604	0.9445	0.5723	0.6518	0.5884	0.3469	0.6625	0.3619	0.6709	0.8247	0.5970
STHSL ^[8]	1.8386	0.6186	1.0107	0.6417	0.9927	0.6809	0.6637	0.6712	0.3653	0.6740	0.3656	0.6541	0.8728	0.6567
STSHN ^[3]	1.5457	0.5376	1.0232	0.5572	1.0262	0.5789	0.7066	0.5704	0.3802	0.7479	0.4018	0.7424	0.8473	0.6224
STGCN-MoE	1.5282	0.6295	0.9883	0.5790	0.9711	0.5576	0.6698	0.5848	0.3459	0.7504	0.3619	0.7258	0.8109	0.6379
ST-MOE ^[37]	1.5453	0.5835	0.9833	0.5817	0.9701	0.5649	0.6609	0.5852	0.3417	0.7483	0.3590	0.7149	0.8100	0.6297
ST-MoGE (ours)	1.4516	0.5223	0.9380	0.5529	0.9206	0.5301	0.6196	0.5524	0.3214	0.6399	0.3453	0.6389	0.7661	0.5728

^③<https://jcst.ict.ac.cn/en/supplement/0ec730a7-5099-41ec-9951-9b2225a50bcc>, Mar. 2026.

Table 2. Overall Performance (MAE and MAPE) of Crime Prediction on the CHI Dataset

Model	Larceny		Battery		Damage		Assault		Fraud		Weapons		Overall	
	MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓
STGCN ^[5]	1.2129	0.5634	0.8199	0.5318	0.6698	0.5430	0.5798	0.5106	0.3862	0.7946	0.2935	0.7444	0.6604	0.6146
GWN ^[5]	1.2522	0.5891	0.8239	0.5598	0.6663	0.5336	0.5835	0.5148	0.4036	0.7293	0.2906	0.8083	0.6700	0.6225
GMAN ^[7]	1.2247	0.5516	0.8189	0.5219	0.6645	0.5237	0.5740	0.5120	0.3917	0.8122	0.2771	0.7277	0.6585	0.6082
MTGNN ^[20]	1.2770	0.5715	0.8299	0.5530	0.6693	0.5266	0.5831	0.5402	0.3999	0.7833	0.2913	0.7990	0.6751	0.6289
AGCRN ^[6]	1.3378	0.5441	0.8284	0.5208	0.6736	0.5140	0.5804	0.5607	0.4091	0.7066	0.2786	0.7310	0.6846	0.5905
MAGNN ^[35]	1.2554	0.5624	0.8333	0.5495	0.6731	0.5216	0.5900	0.5102	0.4006	0.7149	0.2904	0.7795	0.6738	0.6064
FCSTGNN ^[36]	1.2477	0.5661	0.8237	0.5456	0.6755	0.5547	0.5871	0.6131	0.4007	0.8307	0.2826	0.7771	0.6696	0.6479
DeepCrime ^[4]	1.2029	0.5470	0.8082	0.5334	0.6670	0.5280	0.5757	0.5403	0.4128	0.7530	0.2846	0.7267	0.6585	0.6047
STHSL ^[8]	1.3456	0.5607	0.8412	0.5694	0.6971	0.6030	0.6085	0.5370	0.4074	0.7236	0.3117	0.7490	0.7019	0.6237
STSHN ^[3]	1.3133	0.5983	0.9011	0.5271	0.8157	0.7135	0.6310	0.5493	0.4232	0.8213	0.3092	0.9426	0.7323	0.6920
STGCN-MoE	1.2394	0.6154	0.8217	0.5507	0.6784	0.5380	0.5743	0.5171	0.4045	0.7238	0.3067	0.7287	0.6692	0.6123
ST-MOE ^[37]	1.2287	0.5770	0.8278	0.5435	0.6730	0.5290	0.5871	0.5420	0.4018	0.7301	0.3041	0.8050	0.6704	0.6211
ST-MoGE (ours)	1.1816	0.5357	0.8010	0.5174	0.6571	0.5108	0.5589	0.5076	0.3786	0.7038	0.2731	0.7237	0.6417	0.5812

loss. 3) With the design of clustered predictors and a weighting strategy, ST-MoGE could better capture the crime patterns in regions with low crime rates under the imbalanced crime distribution.

- Though all the baseline models can effectively capture spatial-temporal dependencies, their performances are constrained because their architectures are limited to capturing only shared crime patterns. Such limitations make their capacities insufficient to comprehensively preserve heterogeneous crime patterns of different types. Moreover, the captured crime pattern of these models is usually biased to specific categories while performing relatively worse on other categories. For example, in the CHI crime dataset, STGCN and DeepCrime demonstrate remarkable accuracy on high-frequency categories, such as larceny, but perform poorly on low-frequency categories such as robbery. Conversely, AGCRN exhibits better performance on low-frequency categories but struggles with high-frequency ones. This is because mutual contradictions potentially exist between crime patterns of different categories, leading to biases in some categories. Benefiting from the MoE architecture, our ST-MoGE well

handles the spatial-temporal heterogeneity, performing remarkable prediction ability on all the crime categories.

5.3 Ablation Study

To investigate the effectiveness of different components, we conduct the following ablation studies. Note that the bolded data in the tables indicate the best-performing experimental results.

5.3.1 Effect of MGE

Compared with conventional spatial-temporal prediction approaches, a significant difference in our method is the adoption of an MoE architecture, which deploys multiple ST-experts to model spatial-temporal dependencies from different aspects. To evaluate the effectiveness of the MoE architecture, we construct two variants, named “w/o S-Experts” and “w/o U-Expert”, by disabling the category-specific ST-experts and the universal expert, respectively. The results in [Tables 3](#) and [4](#) demonstrate that removing

Table 3. Ablation Study of ST-MoGE Components on the NYC Dataset (MAE Results)

Model	Larceny	Harassment	Assault	Criminal	Indecency	Robbery
W/o U-Experts	1.4780	0.9423	0.9256	0.6340	0.3448	0.3522
W/o S-Expert	1.5028	0.9585	0.9410	0.6234	0.3223	0.3472
W/o CECL	1.4648	0.9428	0.9262	0.6346	0.3228	0.3493
W/o HALR	1.4823	0.9411	0.9263	0.6223	0.3256	0.3536
ST-MoGE	1.4516	0.9380	0.9206	0.6196	0.3214	0.3453

Table 4. Ablation Study of ST-MoGE Components on the CHI Dataset (MAE Results)

Model	Larceny	Battery	Damage	Assault	Fraud	Weapons
W/o U-Experts	1.204 6	0.816 8	0.662 3	0.568 9	0.389 6	0.282 0
W/o S-Expert	1.228 3	0.826 5	0.666 2	0.574 9	0.379 5	0.275 2
W/o CECL	1.201 8	0.808 9	0.659 2	0.559 9	0.383 7	0.275 5
W/o HALR	1.197 9	0.817 2	0.667 4	0.574 2	0.380 7	0.285 0
ST-MoGE	1.181 6	0.801 0	0.657 1	0.558 9	0.378 6	0.273 1

each type of expert leads to a significant reduction in prediction performance. In particular, disabling the category-specific experts significantly impacts the prediction capability for the majority category (e.g., larceny). Without the category-specific experts, the MoE network degenerates into a conventional spatial-temporal graph neural network where all categories share one spatial-temporal pattern, adversely affected by spatial-temporal heterogeneity. Conversely, the removal of the universal expert reduces the prediction accuracy for minority categories (e.g., Indecency, Robbery in the NYC dataset; Fraud, Weapons in the CHI dataset) due to severe sparsity in crime data for these categories, which challenges effective pattern extraction. These experiments underscore the benefits of the MoE architecture, where each type of expert complements the other, positively impacting the overall crime prediction performance.

5.3.2 Effect of CECL

We further conduct experiments by removing the CECL module to evaluate its effectiveness, naming the variant “w/o CECL”. From Tables 3 and 4, we observe that the removal of CECL leads to performance drops across all the categories. This demonstrates the effectiveness of the CECL module and underscores the necessity of decomposing different types of crime patterns.

5.3.3 Effect of HALR

To verify the effectiveness of our proposed HALR, we conduct experiments on the variant “w/o re-weight”, which does not employ the HALR module during the training phase, setting the loss weights of each region cluster equally. The results presented in Tables 3 and 4 show that removing HALR leads to performance degradation, which confirms the effec-

tiveness of HALR.

5.4 Model Robustness

In this subsection, we perform experiments to evaluate the robustness of our ST-MoGE in regions with low crime frequency. We separately evaluate the prediction performance of all crime categories in regions with relatively low crime frequency. For each category, we split regions with less frequent crimes into two groups, at the quantile intervals $(0.00, 0.25]$ and $(0.25, 0.50]$. The evaluation results are presented in Fig.5.

We observe that our ST-MoGE outperforms almost all the other methods in all the cases, demonstrating the robustness of ST-MoGE in regions with extreme situations. The imbalanced spatial distribution of crime occurrences negatively affects spatial-temporal dependencies modeling for neural networks, often leading to biases and insufficient training on low-frequency regions, making it difficult to effectively learn crime patterns in these areas. With the incorporation of our HALR module, ST-MoGE can automatically adjust the importance of regions during the training stage, alleviating the impacts of imbalanced distribution.

Further analysis, including the hyperparameter study and case study, is presented in Appendix A4^④ and Appendix A5^④, respectively.

6 Conclusions

In this paper, we presented Spatial-Temporal Mixture-of-Graph-Experts (ST-MoGE), a framework to address the crime prediction problem. Our framework introduces Mixture-of-Graph-Experts (MGE), which comprehensively captures heterogeneous spatial-temporal dependencies of crime occurrences. Additionally, we incorporated a cross-expert contrastive

^④<https://jcst.ict.ac.cn/en/supplement/0ec730a7-5099-41ec-9951-9b2225a50bcc>, Mar. 2026.

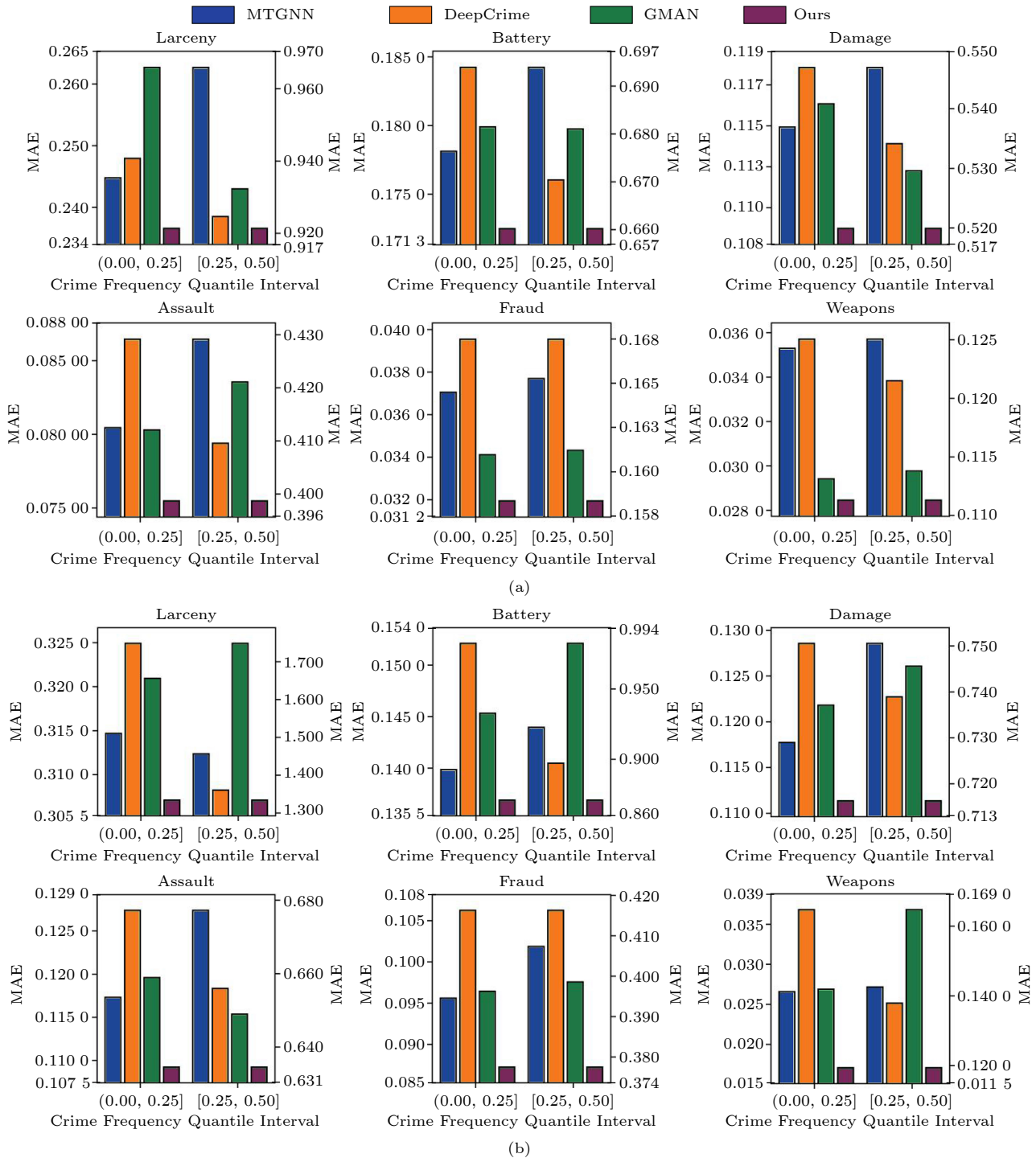


Fig.5. Investigation on the effectiveness of ST-MoGE in low-crime-frequency regions. (a) The NYC dataset. (b) The CHI dataset.

learning (CECL) paradigm to decompose different types of crime patterns, enhancing expert focus and alleviating mutual contradictions among experts. To further improve performance, we employed clustered predictors and the Hierarchical Adaptive Loss Re-Weighting (HALR) module, which mitigates modeling biases and ensures sufficient training for regions

with extreme crime frequencies. When evaluated on real-world datasets, ST-MoGE outperformed existing state-of-the-art methods and demonstrated the efficacy of each key component. Although specifically designed for crime prediction, the foundational principles of our framework have broader applications and could be beneficial in other areas of multi-objective

spatial-temporal prediction, such as urban anomalies prediction.

Conflict of Interest The authors declare that they have no conflict of interest.

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